

# CHERNO BASIRU JALLOW

AI Engineer • Computer Vision Researcher • Startup Co-Founder • Speaker

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## PROFESSIONAL SUMMARY

AI engineer, computer vision researcher, and second-year Information Systems student building practical machine learning systems for healthcare, energy, and education in West Africa. Conducted AI research at the Medical Research Council Unit The Gambia (LSHTM). 1st-place NASA Space Apps Challenge winner, 2nd-place MRCG AI Hackathon, UNICEF Digital Innovation Sprint Top 5 finalist, Deep Learning Indaba 2026 Startup Pitch Top 10 finalist, and ITU Young AI Expert Intern. Co-founder of Voltify, Co-founder and organizer of Hack4Gambia. Recognized as a Top 50 African Innovator by African Innovation Awards. 2x speaker at Google Devfest Banjul, PyCon Senegambia speaker and accepted speaker at PyCon Austria 2026.

## EXPERIENCE

- Young AI Expert Intern • [ITU \(International Telecommunication Union\)](#) Remote | Jul 2026 – Oct 2026
- Developing an internal monitoring and reporting system for ITU-T's Operations & Planning Division, supporting data-driven oversight of AI governance deliverables.
  - Engaging with global telecommunications policy frameworks and contributing technical analysis under ITU-T OPD; part-time at 40% FTE with structured deliverables.
- Data Science / AI Research Intern • [MRC Unit The Gambia at LSHTM](#) Fajara, The Gambia | Mar 2025 – Sep 2025
- Engineered a PyTorch CNN image classifier on a 25,000-image dataset, achieving 80% validation accuracy with custom dataset handling, data validation, and Grad-CAM and saliency-map visualizations
  - Designed and trained a pneumonia detection model using Fastai and transfer learning on 5,000+ chest X-rays, exploring ML viability for low-resource healthcare diagnostics; presented findings at UTG ICT Week 2025 to students, researchers, and clinicians.
  - Implemented a complete neural network from scratch in NumPy: forward propagation, backpropagation via chain rule, gradient descent optimization, and loss minimization.
  - Collaborated in a research-driven environment with biweekly seminars, contributing to reproducibility and scalability discussions in scientific ML workflows under expert supervision.
- Lead AI / ML Intern • [OBENTaS Global](#) Brikama, The Gambia | Feb 2024 – Feb 2025
- Designed and developed a music-preference recommendation system in Python using scikit-learn and pandas to model user tastes through supervised learning; integrated AI features into internal tooling and presented prototypes to executive leadership.
  - Led applied research on e-learning platform design, focusing on user engagement, accessibility, and digital infrastructure constraints in low-resource settings.
  - Represented OBENTaS at The Gambia Trade Fair 2024, engaging government officials, educators, and industry stakeholders; awarded April 2024 Best Team Member for technical output and cross-functional collaboration.
- ICT Club President & Lead Instructor • [Nusrat Senior Secondary School](#) Serrekunda, The Gambia | Jul 2023 – Jul 2024
- Led a 150+ member club, organized inter-class competitions, and raised GMD 10,000+ for educational tech trips; co-founded Coders Hub, a peer-led coding community.
  - Instructed 50+ students weekly in coding fundamentals, web development, MS Office, and graphic design - designing customized lesson plans to advance digital literacy.
- Founder & Content Creator • [CBJtech \(YouTube Channel\)](#) The Gambia | Jan 2023 – Present
- Built a West African AI/ML education platform producing 70+ video tutorials on deep learning, computer vision, research methods, and career development for aspiring developers across the Senegambian region.

## SELECTED PROJECTS

- Voltify - IoT Smart Energy Management System 2025  
Co-Founder | *UNICEF Digital Innovation Sprint 2025 - Top 5 Finalist*
- Engineered an end-to-end IoT platform addressing a critical access gap: millions of African households manage prepaid electricity with no usage visibility, no low-balance warnings, and inaccessible meters requiring 20-digit manual token entry.
  - Built ESP32-based smart meters with real-time current sensing and GSM/LTE connectivity; developed a cross-platform Flutter app with usage tracking, predictive alerts, mobile money integration (Wave/Afrimoney/Comcash), and remote power control.
  - Demonstrated a working prototype before UNICEF, MoBSE, and Government of Japan representatives; advanced to Top 5 from dozens of teams.
- JobMatch AI - AI-Powered Employment & Career Platform 2026  
Co-Founder & ML Engineer | *Deep Learning Indaba 2026 Startup Pitch Competition - Top 10 Finalist*

- Co-founded an AI-powered employment platform connecting job seekers, employers, and training providers across Africa through intelligent opportunity matching and career guidance.
- Developed AI-driven CV analysis and job matching workflows that evaluate candidates' experience, skills, and education against available opportunities while generating explainable match insights.
- Pitched JobMatch AI at the Deep Learning Indaba 2026 Startup Pitch Competition, selected among 10 finalists from 34 applicants; demonstrated the platform to government, industry, and career-development stakeholders.
- Demonstrated the platform at a national career event and to The Gambia's Minister of Higher Education, Research and Technology; product featured on QTV.

Data Pathways to Healthy Cities - Urban Intelligence System

Oct 2025

Data & ML Engineer | NASA Space Apps Challenge Banjul 2025 - 1st Place

- Developed a full-stack geospatial dashboard processing NASA Earth observation satellite data to provide urban planners with actionable insights on city growth, air quality, and vegetation trends.
- Co-engineered end-to-end pipelines transforming raw satellite imagery into interactive visualizations for data-driven, human-centered urban development - winning 1st place among all Gambian teams.

PneumoniaNet - AI-Assisted Childhood Pneumonia Screening

Oct 2025

Lead ML Engineer | MRCG AI Hackathon 2025 - 2nd Place | Presented at OIC Conference Centre

- Designed and trained a lightweight 3-class CNN (Normal / Infiltrate / Consolidation) optimized for CPU inference in resource-constrained clinic settings; implemented a class-weighted loss function to handle severe class imbalance (~12% Consolidation class).
- Integrated Grad-CAM visualizations to surface model decision regions, improving clinician trust and interpretability; presented technical solution and public health impact to judges and clinicians at the OIC Conference Centre.

AWARDS & RECOGNITION

- 1st Place - NASA Space Apps Challenge Banjul 2025
- 2nd Place - MRCG AI Hackathon 2025 | Presented at OIC Conference Centre
- Top 5 Finalist - UNICEF Gambia Digital Innovation Sprint 2025
- Top 10 Finalist - Deep Learning Indaba 2026 Startup Pitch Competition, selected from 34 startups across Africa
- Top 50 African Innovators - African Innovation Awards, 2026
- Full Grant Recipient - Deep Learning Indaba 2026 (Lagos, Nigeria)
- Accepted Speaker - PyCon Austria 2026
- April 2024 Best Team Member - OBENTaS Global

LEADERSHIP & COMMUNITY IMPACT

Organizer & Co-Founder

- [Hack4Gambia](#)

2025-Present

- Annually co-organized a national hackathon bringing together university students, high school participants, mentors, judges, and professionals - coordinating partnerships, judging, sponsorships, and sessions to create a platform for tech innovation.

Lead - Data Science & AI Club / Research Club

- [University of The Gambia \(UTG-ITCA\)](#)

May 2026 – Present

- Leads both the Data Science & AI Club and Research Club at UTG-ITCA, developing membership onboarding, learning pathways, and strategic direction to build practical AI and research capacity among university students. Holds Biweekly sessions to strengthen practical AI and research capacity at UTG.

Instructor/Mentor

- [Deep Learning Series](#)

Jan 2026-Feb 2026

- Organized and delivered biweekly online sessions covering machine learning fundamentals and deep neural networks. (Recordings on my Youtube).

INTERNATIONAL LEADERSHIP & AFFILIATIONS

- Generation Connect Youth Envoy | International Telecommunication Union (ITU) | 2026–2030

FEATURED TALKS & PRESENTATIONS

- Building a Career in AI as a Student in Africa - Google Devfest Banjul 2025
- Beyond the Basics: Python for Deep Learning & Scientific Discovery- PyCon Senegambia 2025
- The Reality of Deepfakes: Mechanics, Threats, and Detection - Google Devfest Banjul 2024
- AI, Innovation, and the Self-Taught Journey (Keynote) - Various tech events 2024,2025 & 2026
- Pneumonia Detection with Fastai: AI for Low-Resource Healthcare - UTG ICT Week 2025 (2025)
- STEM Inspiration & AI Careers in Africa - Nusrat Senior Secondary School (2023 – 2024)

SELECTED RESEARCH & PUBLICATIONS

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- LOWEL AI: An AI-powered system leveraging microscope and sensor data to detect waterborne bacteria in The Gambia. Self-published, 2024.
- Computer Vision Models for Detecting and Classifying Crop Diseases in Gambian Farms - Research Preprint, 2025.

EDUCATION

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B.Sc. Information Systems • [University of The Gambia](#) Faraba, The Gambia | Mar 2025 – 2029

- Relevant coursework: Data Structures & Algorithms, Database Systems, Computer Architecture, Networking, Object-Oriented Programming

TECHNICAL SKILLS

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Languages: Python, JavaScript, Java, C++, HTML/CSS

ML / DL: PyTorch, Fastai, scikit-learn, CNNs, Transfer Learning, Model Evaluation

Computer Vision: OpenCV, Image Classification, Object Detection, Saliency Maps, Interpretability Tools

Data / Research: NumPy, pandas, Jupyter, Scientific Computing, Experimental Design, Reproducibility, Technical Writing

IoT / Edge: ESP32, GSM/LTE, Real-Time Sensor Integration, Edge Deployment

Tools / Platforms: Git, GitHub, Linux, VS Code, Hugging Face

Languages: English (Fluent), Fula (Native), Wolof (Conversational)